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Dear Professor Marc & Selection Team,

I am applying for the EngD on developing an **interactive reverse logistics map** for circular construction. I find this topic very interesting. A map that brings material information, logistics and decisions together can help good materials get reused more often. Today, people often cannot see what is available, where it is, or how reuse compares on cost, CO_2 and circularity. So they choose new materials. I would like to help build a map that makes reuse easier and more reliable.

During my master's studies, my thesis, **Agile Project Management in the Construction Industry**, looked at how Agile can improve teamwork, risk management, decision-making and the use of resources on construction projects. After finishing it, I saw a clear limit: better project delivery and less construction waste only cover one part of sustainability. This led me to a bigger question: what happens to materials when buildings reach the end of their life? When I looked into circular construction, I saw how important reverse logistics is. The main problem is not only whether reuse is technically possible. Information is often missing or spread across many people and companies. Details about material availability, quality, ownership, location, transport and reuse options are often incomplete or not shared. Without clear information, good materials may never return to construction.

I also read earlier research in this field, including Professor Voordijk's work on recovering building elements for reuse. His point is that reuse needs good coordination from the donor building to the target building, not only new technology. This raises clear questions for the map: How can people see which materials are available and in what condition? How can we match supply from a donor building with demand from a target building when timing, quality and cost are uncertain? How can a map help different users compare reuse options, without becoming a tool that nobody uses? These are the questions I want to work on. They also connect to my thesis, where teams had to adapt when plans and conditions changed.

My work experience fits these questions well. At Aug.e I worked on digital twins for smart buildings. We used a digital model to understand and improve how a real building works. The reverse logistics map feels close to that idea, but for materials and how they move. At Hive CPQ I managed large product datasets from manufacturers and turned complex information into simple digital tools that non-technical users could use. At iostring I worked with a construction partner on a circular construction hypothesis: how Agile project management can bring circular practices into real projects. That work stayed at the hypothesis stage. A real (de)construction pilot with partners such as Dura Vermeer and TNO is the best next step. There I can contribute by linking construction knowledge, user needs and digital tools, together with my civil and environmental engineering background and GIS skills. I recently took a career break to care for my newborns. During that time I kept learning about circular construction, reverse logistics and GIS. I am ready to move to the Netherlands for this EngD.

The EngD at the University of Twente is exciting for me because it combines strong study with a real design project in industry. I want to learn reverse-logistics modelling in more depth, use GIS carefully to match donor and target materials, and work in a design-science way: define real use cases with partners, test the map in a pilot, and improve it with the people who will use it. I can contribute energy, a construction view on reuse, and experience turning messy project data into tools that teams can actually use. Building this map is a challenge I am happy to take on, and Twente is a great place to do it because research and industry work closely together.

Thank you for reading my application. I would be happy to talk about how I can contribute to the project.

Yours sincerely,
Pooja Awasare